

Financial Engineering

Graduate · Department of Finance · National Chung Cheng University · Fall 2026

Course Information

Instructor	Zong-Wei Yeh 葉宗瑋
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Class Time	Monday 09:10–12:00
Classroom	管理學院 350
Office Hours	Tuesday 09:00–12:00 (email in advance)

Course Description

A graduate course on the theory and practice of derivative pricing. The first half builds the pricing framework — no-arbitrage and the fundamental theorems, stochastic calculus, and the Black–Scholes model. The second half turns to models beyond Black–Scholes, the volatility surface and its derivatives, and empirical option-return research carried out through paper presentations and a final project.

Course Materials

Main materials:

- Teacher-developed lecture notes and slides, distributed section by section.

Reference books (optional, for additional reading):

- Björk, T., *Arbitrage Theory in Continuous Time*, Oxford.
- Shreve, S. E., *Stochastic Calculus for Finance*, I–II, Springer.
- Duffie, D., *Dynamic Asset Pricing Theory*, Princeton.

Grading

Midterm Exam	30%
Paper Presentation	30%
Final Report	30%
Class Performance	10%

The **final report** is an empirical study that must use **derivatives** data; within that, the instrument and the research question are your own choice. One possible direction, not a requirement, is to use options data such as TAIEX TXO and adapt a study from the Reading List to the Taiwan market.

Weekly Schedule

Week	Date	Topic
1	9/7	Introduction to Financial Engineering; Arbitrage and the Logic of Pricing
2	9/14	Information, Martingales, and Multi-Period Markets
3	9/21	Brownian Motion and Itô Calculus
4	9/28	教師節（放假）
5	10/5	Brownian Motion and Itô Calculus
6	10/12	The Black–Scholes World
7	10/19	Computational Methods
8	10/26	光復節（放假）
9	11/2	Midterm Exam (S1–S4)
10	11/9	Implied Volatility and the Smile
11	11/16	Beyond Black–Scholes
12	11/23	Invited Speech
13	11/30	Paper Presentation I
14	12/7	Paper Presentation II
15	12/14	Volatility Derivatives
16	12/21	Final Report

Reading List

Papers for the paper presentations (Weeks 13–14) and the final report:

- Broadie, M., Chernov, M., & Johannes, M. (2009). Understanding index option returns. *The Review of Financial Studies*, 22(11), 4493–4529.
- Coval, J. D., & Shumway, T. (2001). Expected option returns. *The Journal of Finance*, 56(3), 983–1009.
- Goyal, A., & Saretto, A. (2009). Cross-section of option returns and volatility. *Journal of Financial Economics*, 94(2), 310–326.
- Zhan, X., Han, B., Cao, J., & Tong, Q. (2022). Option return predictability. *The Review of Financial Studies*, 35(3), 1394–1442.
- Bali, T. G., Chabi-Yo, F., & Murray, S. (2022). A factor model for stock returns based on option prices. SSRN 3487947.
- Neuhierl, A., Tang, X., Varneskov, R. T., & Zhou, G. (2025). Do option characteristics predict the underlying stock returns in the cross-section? *Management Science*.
- Kelly, B. T., Kuznetsov, B., Malamud, S., & Xu, T. A. (2023). Deep learning from implied volatility surfaces. *Swiss Finance Institute Research Paper*, (23–60).